

PHYSICS**Paper 1 (Theory)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 10 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

- | | | |
|-----------------------------|--------------------|---|
| 1. Planck's constant | h | $= 6.6 \times 10^{-34} \text{ J s}$ |
| 2. Speed of light in vacuum | c | $= 3 \times 10^8 \text{ m s}^{-1}$ |
| 3. Charge of a proton | e | $= 1.6 \times 10^{-19} \text{ C}$ |
| 4. Permittivity of vacuum | ϵ_0 | $= 8.85 \times 10^{-12} \text{ F m}^{-1}$ |
| 5. | $1/4\pi\epsilon_0$ | $= 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$ |
| 6. Permeability of vacuum | μ_0 | $= 4\pi \times 10^{-7} \text{ H m}^{-1}$ |
| 7. One atomic mass unit | 1 u | $= 1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$ |
| 8. One electron volt | 1 eV | $= 1.6 \times 10^{-19} \text{ J}$ |

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

- (A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.
- (i) A closed surface S encloses three point charges $+4\ \mu\text{C}$, $-6\ \mu\text{C}$ and $+8\ \mu\text{C}$. A fourth charge of $+5\ \mu\text{C}$ lies just outside S . The electric flux through S is: (*Apply*) [1]
- (a) $6.78 \times 10^5\ \text{N m}^2\ \text{C}^{-1}$
(b) $1.24 \times 10^6\ \text{N m}^2\ \text{C}^{-1}$
(c) $1.02 \times 10^6\ \text{N m}^2\ \text{C}^{-1}$
(d) zero, because the charges outside cancel those inside
- (ii) Select the odd one out from the following materials. (*Analyse*) [1]
- (a) Copper
(b) Bismuth
(c) Water
(d) Aluminium
- (iii) A circular coil of N turns and radius R produces a field B_0 at its centre. The same wire is unwound and rewound into a coil of radius $R/2$ and the same current is passed. The field at the new centre is: (*Apply*) [1]
- (a) $B_0/2$
(b) $2B_0$
(c) $4B_0$
(d) $B_0/4$
- (iv) A proton and an α -particle are accelerated from rest through the same potential difference. The ratio of their de Broglie wavelengths $\lambda_p : \lambda_\alpha$ is: (*Apply*) [1]
- (a) 1 : 1
(b) 2 : 1
(c) $2\sqrt{2} : 1$
(d) 1 : $2\sqrt{2}$
- (v) Identify the correct statements about a p-type semiconductor. (*Understand*) [1]
- (P) Holes are the majority charge carriers.
(Q) It is obtained by doping silicon with a pentavalent impurity.
(R) It is obtained by doping silicon with a trivalent impurity such as boron.
(S) It carries a net positive charge.
- (a) Both (P) and (Q)
(b) Both (P) and (R)
(c) Both (Q) and (S)
(d) Both (R) and (S)

- (vi) Match the entries in Column I with those in Column II. (*Recall*) [1]

Column I	Column II
1. Optical fibre	A. Total internal reflection
2. Spreading of light at a narrow slit	B. Dispersion
3. Bright and dark bands in Young's experiment	C. Diffraction
4. Splitting of white light by a prism	D. Interference

- (a) 1-A, 2-C, 3-D, 4-B
 (b) 1-A, 2-D, 3-C, 4-B
 (c) 1-B, 2-C, 3-D, 4-A
 (d) 1-C, 2-A, 3-B, 4-D
- (vii) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]

Assertion: A convex glass lens ($n = 1.5$) has a longer focal length when it is immersed in water ($n = 1.33$) than in air.

Reason: The power of a lens depends on the refractive index of the lens material relative to the surrounding medium. (*Analyse*)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.
- (B) Answer the following questions briefly:
- (i) How much work is done in moving a charge of $2 \mu\text{C}$ through 5 cm along an equipotential surface? (*Recall*) [1]
- (ii) What is meant by motional emf? (*Recall*) [1]
- (iii) Fill in the blank with an appropriate term from the bracket.
 The refractive index of glass for violet light is _____ that for red light. (greater than, less than, equal to) (*Recall*) [1]
- (iv) What is the phase difference between any two points on the same wavefront? (*Recall*) [1]
- (v) Calculate the momentum of a photon of energy $3.3 \times 10^{-19} \text{ J}$. (*Apply*) [1]
- (vi) What is the minimum energy of a γ -ray photon that can create an electron-positron pair? (*Recall*) [1]
- (vii) What happens to the width of the depletion region of a p-n junction when it is reverse biased? (*Recall*) [1]

SECTION B – 14 MARKS

Question 2

- (i) Three capacitors of capacitances $2\ \mu\text{F}$, $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are connected in series across a $12\ \text{V}$ battery. Calculate (a) the equivalent capacitance and (b) the potential difference across the $6\ \mu\text{F}$ capacitor. (*Apply*) [2]

OR

- (ii) An infinitely long straight wire with a uniform linear charge density produces an electric field of $1.8 \times 10^5\ \text{N C}^{-1}$ at a distance of $5\ \text{cm}$ from it. Calculate the linear charge density. What will the field be at $10\ \text{cm}$? (*Apply*) [2]

Question 3

- Draw a labelled circuit diagram of the arrangement used in the laboratory to find the internal resistance of a cell using a potentiometer. Write the formula used. (*Understand*) [2]

Question 4

- (i) A copper wire of cross-sectional area $2.0\ \text{mm}^2$ carries a current of $4.0\ \text{A}$. The number density of free electrons in copper is $8.5 \times 10^{28}\ \text{m}^{-3}$. Calculate the drift speed of the electrons. How would the drift speed change if the wire were replaced by a wire of the same material but of twice the radius, carrying the same current? (*Apply*) [2]

OR

- (ii) Figure 1 shows a junction O in a circuit. Find the current I . Which conservation law is used? (*Apply*) [2]

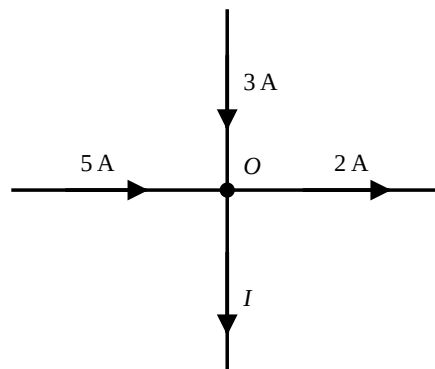


Figure 1

Question 5

- A galvanometer of resistance $40\ \Omega$ gives full-scale deflection for a current of $5\ \text{mA}$. How will you convert it into a voltmeter of range $0\text{--}10\ \text{V}$? Draw the arrangement. (*Apply*) [2]

Question 6

- Two thin lenses of powers $+5\ \text{D}$ and $-2.5\ \text{D}$ are placed in contact. Find the focal length and nature of the combination. Where should an object be placed so that the combination forms a real image of the same size? (*Apply*) [2]

Question 7

- (i) Name the electromagnetic waves used (a) in water purifiers to kill germs, and (b) in the remote control of a television set. (*Recall*) [1]
- (ii) An electromagnetic wave has a wavelength of 10^{-10} m. Name it and state one of its uses. (*Recall*) [1]

Question 8

A laser emits light of wavelength 600 nm with a power of 6.6 mW. Calculate (a) the energy of each photon in eV, and (b) the number of photons emitted per second. (*Apply*) [2]

SECTION C – 27 MARKS**Question 9**

- (i) Obtain an expression for the electric field intensity at a point on the axial line of an electric dipole of dipole moment $p = q \times 2l$, at a distance r from its centre. Write the expression for a short dipole. (*Understand*) [3]

OR

- (ii) Three capacitors of capacitances C_1 , C_2 and C_3 are connected in series across a potential difference V . With the help of a labelled diagram, obtain an expression for their equivalent capacitance. (*Understand*) [3]

Question 10

- (i) In the network shown in Figure 2, the cells have negligible internal resistance. Using Kirchhoff's laws, calculate the currents I_1 , I_2 and I_3 . Which cell is being charged? (*Apply*) [3]

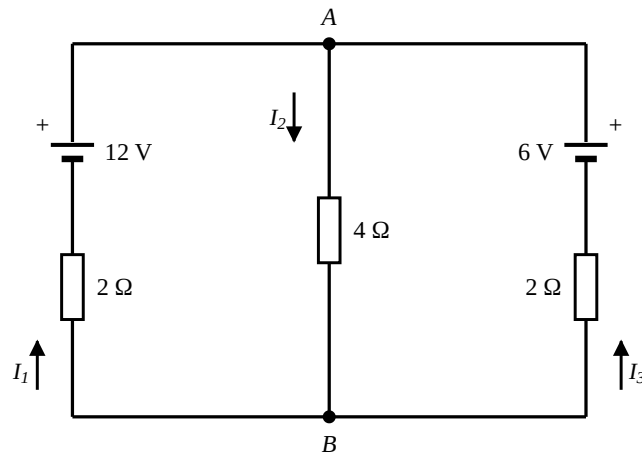


Figure 2

OR

- (ii) In the metre bridge shown in Figure 3, the null point D is at 40 cm from end A when an unknown resistance X is in the left gap and $12\ \Omega$ in the right gap. (a) Find X . (b) What resistance must be connected in parallel with the $12\ \Omega$ so that the null point moves to the middle of the wire? (*Apply*) [3]

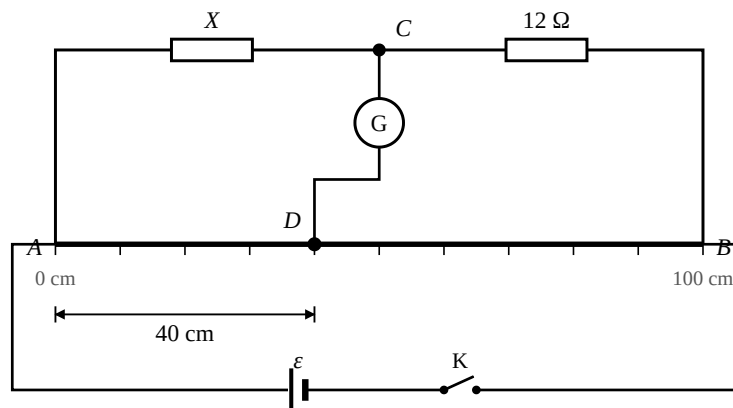


Figure 3

Question 11

- State the Biot–Savart law. Using it, obtain an expression for the magnetic field at the centre of a circular coil of N turns and radius R carrying a current I . (*Understand*) [3]

Question 12

- (i) Two long, straight, parallel wires 10 cm apart carry currents of 5 A and 8 A in the **same** direction. Calculate the force per metre length on each wire and state whether it is attractive or repulsive. (*Apply*) [2]

- (ii) The magnetic susceptibility of a paramagnetic salt is 3.6×10^{-4} at 27°C . What will it be at -73°C ? (*Apply*) [1]

Question 13

A physics teacher set up Young's double slit experiment as shown in Figure 4 and asked the following questions. State whether the answer of the student is correct or incorrect. Correct the incorrect response with a brief explanation. (*Evaluate*)

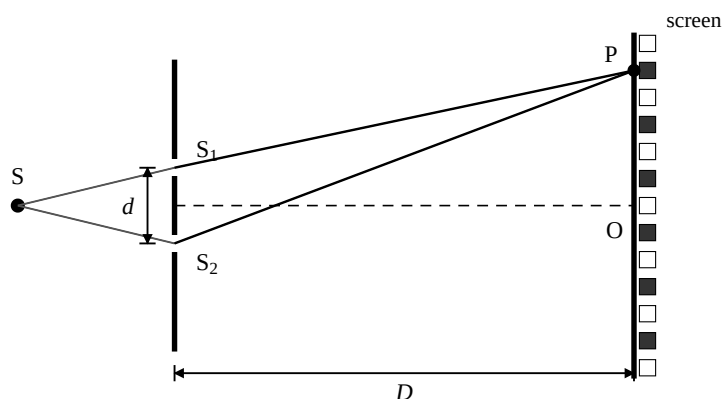


Figure 4

- (i) **Teacher:** If the whole apparatus is immersed in water ($n = 4/3$), what happens to the fringe width? [1]
Student: The fringe width increases.
- (ii) **Teacher:** If slit S_2 is covered and slit S_1 is made narrower, what happens to the central maximum of the diffraction pattern on the screen? [1]
Student: It becomes wider.
- (iii) **Teacher:** If the two slits are lit by two separate sodium lamps instead of the single source S, what will be seen? [1]
Student: Interference fringes of the same width.

Question 14

Using Huygens' principle, draw the refracted wavefront when a plane wavefront is incident on a plane surface separating a rarer medium from a denser medium. Hence prove Snell's law of refraction. (*Understand*) [3]

Question 15

- (i) A thin equiconvex lens made of glass of refractive index 1.5 has a focal length of 20 cm in air. Calculate the radius of curvature of its surfaces. What will its focal length be when it is immersed in a liquid of refractive index 1.25? (*Apply*) [2]
- (ii) A ray of light is incident normally on one face of a glass prism of angle 30° and refractive index $\sqrt{2}$. Find the angle of deviation. (*Apply*) [1]

Question 16

- (i) Draw a labelled ray diagram of an astronomical (refracting) telescope in normal adjustment. (*Understand*) [2]
- (ii) Write an expression for its magnifying power, and state why the objective is given a large aperture. (*Recall*) [1]

Question 17

Figure 5 shows the variation of stopping potential V_s with the frequency ν of the incident radiation for two metals A and B.

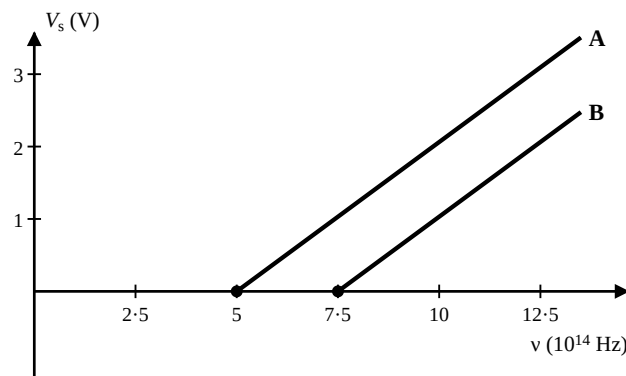


Figure 5

- (i) Which metal has the greater work function? Give a reason. (*Analyse*) [1]
- (ii) Why are the two graphs parallel? What does the slope represent? (*Understand*) [1]
- (iii) Calculate the work function of metal A in eV. (*Apply*) [1]

SECTION D – 15 MARKS

Question 18

- (i) A series LCR circuit has $R = 120 \, \Omega$, $L = 0.2 \, \text{H}$ and $C = 40 \, \mu\text{F}$. It is connected to a source of emf $e = 260\sqrt{2} \sin 500t$ volt.
 - (a) Calculate the inductive and capacitive reactances and the impedance. (*Apply*) [1]
 - (b) Find the rms current and the peak current. (*Apply*) [1]
 - (c) Find the phase angle between the voltage and the current. Which leads? (*Analyse*) [1]
 - (d) Calculate the average power consumed and the power factor. (*Apply*) [1]

- (e) What value of capacitance, used with the same inductor, would make the circuit resonate at this frequency? (*Apply*) [1]

OR

- (ii)(a) Figure 6 shows the variation of the voltage V across an electrical device and the current I through it with time t .

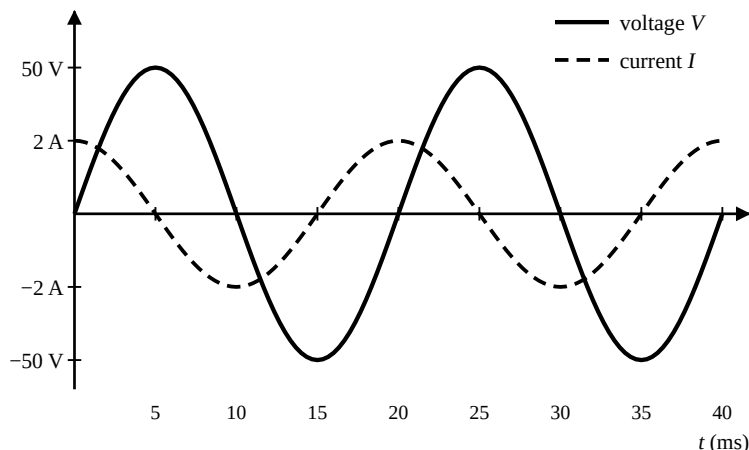


Figure 6

- (1) Identify the device (resistor, inductor or capacitor). Give a reason. (*Analyse*) [1]
- (2) Find the frequency of the supply. (*Apply*) [1]
- (3) Calculate its reactance and hence its capacitance / inductance. (*Apply*) [1]
- (b) A step-down transformer reduces 220 V to 11 V. Its primary has 2000 turns. Find the number of turns in the secondary. If the secondary delivers 4 A to a load, what is the primary current, assuming the transformer is ideal? (*Apply*) [2]

Question 19

- (i)(a) Using Bohr's postulates, obtain an expression for the radius of the n^{th} orbit of the electron in a hydrogen atom and show that $r_n \propto n^2$. (*Apply*) [3]
- (b) Draw a graph showing the variation of binding energy per nucleon with mass number. Mark on it the regions where fission and fusion release energy. (*Understand*) [2]

OR

- (ii)(a) Name one material used as the **moderator** and one used for the **control rods** in a nuclear reactor, and state the function of each. (*Recall*) [2]
- (b) The radius of the first Bohr orbit of hydrogen is 0.053 nm. What is the radius of the second orbit? (*Apply*) [1]
- (c) Why does nuclear fusion need a very high temperature? (*Understand*) [1]
- (d) What are isobars? Give one example. (*Recall*) [1]

Question 20

A school installs **smart solar street lamps**. During the day a panel of silicon solar cells charges a battery. At dusk a **photodiode**, connected in reverse bias, senses the fall in daylight and switches on a lamp made of red **LEDs** of wavelength 662 nm. The solar cell works without any external bias; the LEDs are forward biased.

- (i) Why is the photodiode operated in reverse bias rather than forward bias? (*Understand*) [1]
- (ii) Draw the I–V characteristic of a solar cell and mark the open-circuit voltage and the short-circuit current on it. (*Understand*) [1]
- (iii) Estimate the band gap (in eV) of the semiconductor used in the LEDs. (*Apply*) [1]
- (iv) Name the three basic processes by which a solar cell produces an emf. (*Recall*) [1]
- (v) State one advantage of an LED lamp over an incandescent lamp. (*Recall*) [1]

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